THE HENSELIZATION OF A PAIR IS FLAT AND DOES NOT CHANGE THE QUOTIENTS BY POWERS OF THE IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Construction of the henselization as a filtered colimit of etale algebras inducing an isomorphism modulo the ideal. Lemma \ref{lemma-henselization}, cited at proof lines 3095-3098. Provides the colimit presentation used for both statements.
- Etale ring maps are flat and flatness is preserved by filtered colimits. Algebra, Lemma \ref{algebra-lemma-etale} and Algebra, Lemma \ref{algebra-lemma-colimit-flat}, cited at proof lines 3098-3100. Give flatness of the henselization.
- Flat maps that induce an isomorphism modulo the ideal induce isomorphisms on the quotients by all powers. Algebra, Lemma \ref{algebra-lemma-lift-basis}, cited at proof lines 3101-3103. Upgrades the modulo-the-ideal isomorphism to the quotients by powers.

Source reference. Original lemma 0AGU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13207–13207. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0A02: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12620–12620. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00U2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1096–1096. Navigation pointer to a statement the

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Source reference. Original lemma 05UT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7527–7527. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 051H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6535–6535. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let (A, I) be a pair. Let (A^h, I^h) be as in Lemma 0A02. Then $A \to A^h$ is flat, $I^h = IA^h$ and $A/I^n \to A^h/I^nA^h$ is an isomorphism for all n.*

Proof. In the proof of Lemma 0A02 we have seen that A^h is a filtered colimit of étale A-algebras B such that $A/I \to B/IB$ is an isomorphism and we have seen that $I^h = IA^h$. As an étale ring map is flat (Algebra, Lemma 00U2) we conclude that $A \to A^h$ is flat by Algebra, Lemma 05UT. Since each $A \to B$ is flat we find that the maps $A/I^n \to B/I^nB$ are isomorphisms as well (for example by Algebra, Lemma 051H). Taking the colimit we find that $A/I^n = A^h/I^nA^h$ as desired. □